

QUANTIZATION OF GAUGE THEORIES WITH WEYL FERMIONS

O. BABELON, F.A. SCHAPOSNIK¹ and C.M. VIALLET

*Laboratoire de Physique Théorique et des Hautes Energies²,
Tour 16, 1er Etage, 4, Place Jussieu, F-75230 Paris Cedex 05, France*

Received 17 June 1986

It is shown that, in the covariant lagrangian formalism, a proper treatment of the gauge degrees of freedom in ordinary Yang–Mills theory with Weyl fermions uncovers the presence of the Wess–Zumino action: the group of gauge transformations gets quantized and the anomaly is absorbed.

1. Introduction. The proposal we present here is the natural prolongation of the conclusions of ref. [1] on the geometrical meaning of the method of quantization for gauge theories [2] to the case where Weyl fermions are present, quite in the spirit of ref. [3]. One of the motivations is the recent interest in the quantization of anomalous gauge theories, revived by ref. [4] (see also ref. [5]).

Indeed it was shown in ref. [1] that, in the lagrangian formalism, the functional integral (generating functional Z) should be considered *as an integral over the whole space of connections, rather than as an integral over the orbit space*.

When Dirac fermions are present (no axial couplings), there is no harm in looking at Z as an integral over the orbit space. On the contrary, when Weyl fermions are present – and because of their ability to create gauge non-invariant interactions [6] – this is not the case anymore, and one has to reconsider the standard [2] procedure of quantization.

The outcome is that the gauge transformations acquire the status of a physical field, which enters a non-trivial effective action, the Wess–Zumino action [7] being part of it. Notice that we do not have to add this term by hand, since it emerges naturally in the process of quantization, nor do we have to introduce an additional field as in ref. [4].

The same phenomenon happens with the Liouville action in the quantization of strings [3].

2. The generating functional. We start from the path integral version of the generating functional Z :

$$Z = \int \mathcal{D}A \mathcal{D}\bar{\psi} \mathcal{D}\psi \exp\left(i \int \mathcal{L}(A, \psi)\right),$$

where

$$\mathcal{L}(A, \psi) = \bar{\psi}(i\partial + A)\psi + \frac{1}{4} \text{tr} F_{\mu\nu}^2 = L_F + L_{YM}.$$

The fermions here are Weyl fermions: $(\psi = \frac{1}{2}(1 - \gamma_5)\psi)$.

If γ is an element of the group of gauge transformations $\mathcal{G}$, we denote its action as follows:

$A\gamma$ is the usual gauge transformed of A by γ (i.e.: $(A\gamma)_\mu = \gamma^{-1}A_\mu\gamma + \gamma^{-1}\partial_\mu\gamma$), and $\gamma^{-1}\psi$ is the gauge transform of the fermion in its unitary representation.

$\mathcal{L}$ is invariant under $\mathcal{G}$.

Suppose we choose a section σ in the space $\mathcal{G}$ of all gauge potentials: a general potential A is gauge related by a (unique) element g to some point σ on this section. Thus the choice of σ is a coordinization of the space $\mathcal{G}$ (see fig. 1).

The functional Z can be written

$$Z = \int \mathcal{D}v(\sigma) \rho(\sigma) \mathcal{D}g \mathcal{D}\bar{\psi} \mathcal{D}\psi \exp[iS(\sigma, g, \psi)],$$

where $-S(\sigma, g, \psi) = \int \mathcal{L}(A, \psi)$ with $A = \sigma g$.

¹ F.A.S. is partially supported by Facultad de Ciencias Exactas, UNLP and CIC, Buenos Aires, Argentina.

² Unité Associée CNRS UA280.

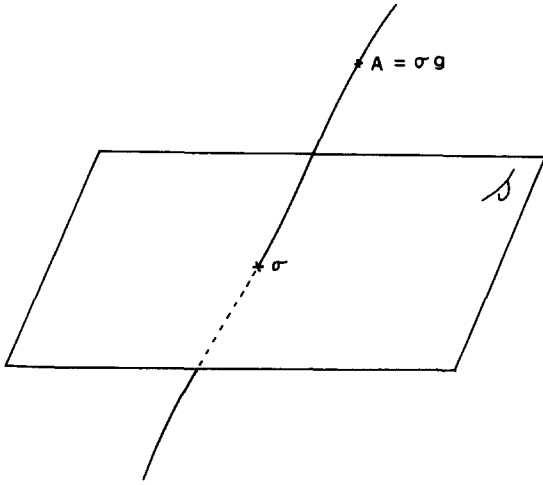


Fig. 1. A section in the space of gauge potentials and the fibre through σ .

(i) $\mathcal{D}v(\sigma)$ is the volume element on the orbit space expressed in the coordinate system given by σ , i.e.: $\mathcal{D}v(\sigma) = \mathcal{D}\sigma \sqrt{\det \Gamma}$ if Γ is the natural metric on the orbit space [8].

(ii) $\rho(\sigma)\mathcal{D}g$ is the volume element on the orbit through σ and decomposes in $\mathcal{D}g$, which is the volume element on the group of gauge transformations (supposed to be bi-invariant), and $\rho(\sigma)$ which is the scale of the orbit. Of course $\int \mathcal{D}g$ is the infinite volume of the group of gauge transformations. The link between $\mathcal{D}v(\sigma)$, $\rho(\sigma)$ and the Faddeev-Popov determinant Δ corresponding to the choice of gauge section σ is very simple [1]:

$$\Delta(\sigma) = \rho(\sigma) \sqrt{\det \Gamma(\sigma)}.$$

(iii) The measure over Weyl fermions is $\mathcal{D}\bar{\psi} \mathcal{D}\psi \exp[i \int L_F(A, \psi)]$. It is not invariant by gauge transformation. Actually it transforms under $\mathcal{g}$ by a jacobian as follows [9]:

If the determinant $\det_L(i\hat{D} + \not{A})$ over left-handed fermions is defined for example as in ref. [10] with ζ -function regularization through the operator $\Delta(A) = i\hat{D} + \not{A} \frac{1}{2}(1 - \gamma_5)$ acting on Dirac fermions:

$$\det_L(i\hat{D} + \not{A}) = \exp[-d\zeta(s, \Delta(A))/ds|_{s=0}],$$

and if γ is an element of $\mathcal{g}$, and $\psi' = \gamma^{-1}\psi$, then

the measure over fermions transforms by:

$$\begin{aligned} \mathcal{D}\bar{\psi}' \mathcal{D}\psi' \exp\left(i \int L_F(A, \psi')\right) \\ = \mathcal{D}\bar{\psi} \mathcal{D}\psi \exp\left(i \int L_F(A\gamma^{-1}, \psi)\right) \mathcal{J}(A\gamma^{-1}, \gamma) \end{aligned}$$

with

$$\mathcal{J}(B, \gamma) = \det_L(i\hat{D} + \not{C}) / \det_L(i\hat{D} + \not{B}) \quad C = B\gamma. \quad (1)$$

With these conventions, Z may be written, by the change of fermionic variables $\psi \rightarrow g^{-1}\psi$:

$$\begin{aligned} Z = \int \mathcal{D}v(\sigma) \rho(\sigma) \mathcal{D}g \mathcal{D}\bar{\psi} \mathcal{D}\psi \mathcal{J}(\sigma, g) \\ \times \exp\left(i \int \mathcal{L}(\sigma, \psi)\right). \end{aligned} \quad (2)$$

Would the fermions be Dirac fermions, then one would have $\mathcal{J}(\sigma, g) = 1$, and the integral over $\mathcal{g}$ would just produce an infinite factor $\int \mathcal{D}g$ for Z , as in ref. [2]. But this is not true for Weyl fermions, and the integral over $\mathcal{g}$ cannot be factorized.

Claim. Z is independent on the choice of gauge section.

This is a direct consequence of the fact that Z may be computed in any system of coordinates, but we will show how the integration over $\mathcal{g}$ wipes out the non-invariance created by the integration over the fermions.

Suppose that σ' is another gauge section, parametrized by σ' , where σ' is gauge related to σ by some γ : $\sigma' = \sigma\gamma$ (of course γ depends on σ).

From eq. (1), we have

$$\mathcal{J}(\sigma, g) = \mathcal{J}(\sigma, \gamma) \mathcal{J}(\sigma\gamma, \gamma^{-1}g). \quad (3)$$

Moreover, $\mathcal{D}v(\sigma') = \mathcal{D}v(\sigma)$ since $\mathcal{D}v(\sigma)$ is an invariant volume element on the orbit space [1], and $\rho(\sigma') = \rho(\sigma)$ since $\rho(\sigma)$ is a function of the orbit through σ and not of σ .

We have to compare Z given by eq. (2) with

$$Z' = \int \mathcal{D}v(\sigma') \rho(\sigma') \mathcal{D}g' \mathcal{D}\bar{\psi}' \mathcal{D}\psi \mathcal{J}(\sigma', g') \\ \times \exp\left(i \int \mathcal{L}(\sigma', \psi')\right).$$

We will compare

$$I = \mathcal{D}\bar{\psi} \mathcal{D}\psi \int \mathcal{D}g \mathcal{J}(\sigma, g) \exp\left(i \int \mathcal{L}(\sigma, \psi)\right)$$

and

$$I' = \mathcal{D}\bar{\psi}' \mathcal{D}\psi' \int \mathcal{D}g' \mathcal{J}(\sigma', g') \\ \times \exp\left(i \int \mathcal{L}(\sigma', \psi')\right).$$

In I' we choose $g' = \gamma^{-1}g$ as an integration variable over the group. Then

$$I' = \mathcal{D}\bar{\psi}' \mathcal{D}\psi' \int \mathcal{D}g \mathcal{J}(\sigma\gamma, \gamma^{-1}g) \\ \times \exp\left(i \int \mathcal{L}(\sigma, \gamma\psi')\right).$$

Setting $\psi = \gamma^{-1}\psi'$ we get

$$I' = \mathcal{J}(\sigma, \gamma) \mathcal{D}\bar{\psi} \mathcal{D}\psi \\ \times \int \mathcal{D}g \mathcal{J}(\sigma\gamma, \gamma^{-1}g) \exp\left(i \int \mathcal{L}(\sigma, \psi)\right),$$

which proves the claim by using eq. (3). $\square$

Corollary. We have shown that $\int \mathcal{D}g \mathcal{D}\bar{\psi} \mathcal{D}\psi \times \mathcal{J}(\sigma, g) \exp(i \int \mathcal{L}(\sigma, \psi))$ is a function of the gauge equivalence class of σ (orbit through σ), and not of the precise choice of σ on the orbit.

If we want to take eq. (2) as a starting point for perturbation theory, we have to implement the gauge condition on A through the standard Faddeev–Popov procedure. *The independence on the choice of gauge condition then translates into BRS invariance* [11].

Notice that $\log(J(\sigma, g))$ in eq. (2) contains the Wess–Zumino action.

3. An example: the two-dimensional theory. We use Minkowsky space–time (x^0, x^1) with metric

$$g_{00} = 1 \text{ and } g_{11} = -1.$$

$$\{\gamma_\mu, \gamma_\nu\} = 2g_{\mu\nu}, \quad \gamma_5 = \gamma_0\gamma_1, \quad \gamma^\mu\gamma_5 = \epsilon^{\mu\nu}\gamma_\nu.$$

$$x_\pm = x^0 \pm x^1, \quad \partial_\pm = \partial_0 \pm \partial_1.$$

We may write

$$A_+ = iU^{-1} \partial_+ U, \quad A_- = iV \partial_- V^{-1},$$

where U and V are group-valued.

For a group valued function a we define

$$\phi(a) = \frac{1}{16\pi} \text{tr} \int d^2x \partial_\mu a^{-1} \partial^\mu a - \Gamma(a),$$

with

$$\Gamma(a) = -\frac{i}{8\pi} \text{tr} \int d^2x \\ \times \int_0^1 dt \epsilon^{\mu\nu} a_t^{-1} \partial_t a_t a_t^{-1} \partial_\mu a_t a_t^{-1} \partial_\nu a_t,$$

if a_t is an interpolating one-parameter family ($0 \leq t \leq 1$) such that $a_0 = \mathbf{1}$ and $a_1 = a$.

Fermion determinants may be computed exactly [12] if defined as is explained above:

$$\det_L(i\not{D} + \not{A}) = \exp[i\phi(U)].$$

We then have

$$Z = \int \mathcal{D}A \exp\left(i \int L_{YM} + i\phi(U)\right).$$

Suppose we choose the gauge “condition” $A_- = 0$. The generic parameter on the gauge section is $\sigma = UV$, and $g = V^{-1}$.

From the property of ϕ :

$$\phi(\alpha \cdot \beta) = \phi(\alpha) + \phi(\beta)$$

$$- \frac{1}{16\pi} \text{tr} \int (\alpha^{-1} \partial_+ \alpha) (\beta \partial_- \beta^{-1}) d^2x,$$

we have:

$$Z = \int \mathcal{D}v(\sigma) \rho(\sigma) \mathcal{D}g \\ \times \exp\left(i \int L_{YM} + i\phi(\sigma) + i\phi(g)\right. \\ \left. - \frac{1}{16\pi} \text{tr} \int (\sigma^{-1} \partial_+ \sigma) (g \partial_- g^{-1})\right).$$

We might as well re-introduce fermions and

replace $\exp[i\phi(\sigma)]$ by $\int \mathcal{D}\bar{\chi} \mathcal{D}\chi \exp(i \int L_F(\sigma, \chi))$ thus

$$Z = \int \mathcal{D}v(\sigma) \rho(\sigma) \mathcal{D}g \mathcal{D}\bar{\chi} \mathcal{D}\chi \\ \times \exp\left(i \int \mathcal{L}(\sigma, \chi) + i\phi(g) - \frac{1}{16\pi} \int (\sigma^{-1} \partial_+ \sigma)(g \partial_- g^{-1})\right),$$

which is just eq. (2) as expected. It clearly exhibits the non-trivial dependence of the quantum theory on g .

4. Conclusion. In the light of the results of ref. [3] and those presented here, it is reasonable to believe that the phenomenon we describe may happen with any classical symmetry: integration over the symmetry degrees of freedom takes care of the anomaly.

To have a complete comprehension of this phenomenon, one will have to establish the precise connection between the lagrangian approach taken here and the hamiltonian approach [4], and also the incidence of the choice of regularization on the final theory.

We finally have to say that, in a realistic model including both left- and right-handed fermions in various representations, not all gauge transformations play the same role in the production of anomalies. There might be a subgroup $\mathcal{H}$ of $\mathcal{G}$ without anomalies and only a residual field in $\mathcal{G}/\mathcal{H}$ becomes physical.

References

- [1] O. Babelon and C.M. Viallet, Phys. Lett. B 85 (1979) 246.
- [2] L.D. Faddeev and V.N. Popov, Phys. Lett. B 25 (1967) 29.
- [3] A.M. Polyakov, Phys. Lett. B 103 (1981) 207.
- [4] L.D. Faddeev, Phys. Lett. B 145 (1984) 81;
L.D. Faddeev and S.L. Shatashvili, Phys. Lett. B 167 (1986) 225.
- [5] R. Jackiw and R. Rajaraman, Phys. Rev. Lett. 54 (1985) 1219;
R. Rajaraman, Phys. Lett. B 154 (1985) 305;
R. Rajaraman, Phys. Lett. B 162 (1985) 148;
I.M. Singer, in: Elie Cartan et les mathématiques d'aujourd'hui, Asterisque (1985);
A.J. Niemi and G.W. Semenoff, Phys. Rev. Lett. 56 (1986) 1019;
L. Alvarez-Gaumé, S. Della Pietra and V. Della Pietra, Phys. Lett. B 166 (1986) 177;
J.G. Halliday, E. Rabinovici, A. Schwimmer and M. Chanowitz, Nucl. Phys. B 268 (1986) 413;
M. Chanowitz, Phys. Lett. B 171 (1986) 280;
E. d'Hoker and E. Farhi, Nucl. Phys. B 248 (1984) 59.
- [6] S. Adler, Brandels Summer School (1970), eds. S. Deser, M. Grisaru and D. Pendleton (MIT Press, Cambridge, MA).
- [7] J. Wess and B. Zumino, Phys. Lett. B 37 (1971) 95.
- [8] O. Babelon and C.M. Viallet, Commun. Math. Phys. 81 (1981) 515;
C.M. Viallet, Lectures XXII Karpacz Winter School of Theoretical Physics (1986), ed. A. Jadczyk (World Scientific, Singapore), to be published.
- [9] K. Fujikawa, Phys. Rev. D21 (1980) 2848.
- [10] L. Alvarez-Gaumé and P. Ginsparg, Nucl. Phys. B 243 (1984) 449.
- [11] C. Becchi, A. Rouet and R. Stora, Ann. Phys. (NY) 98 (1976) 287.
- [12] R.E. Gamboa-Saravi and M.A. Muschietti, F.A. Schaposnik and J.E. Solomin, Ann. Phys. (NY) 157 (1984) 360;
A.M. Polyakov and P.B. Wiegmann, Phys. Lett. B 141 (1984) 223.